

CoGenesis: A Framework Collaborating Large and Small Language Models for Secure Context-Aware Instruction Following

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Abstract

With the advancement of language models (LMs), their exposure to private data is increasingly inevitable, and their deployment (especially for smaller ones) on personal devices, such as PCs and smartphones, has become a prevailing trend. In contexts laden with user information, enabling models to both safeguard user privacy and execute commands efficiently emerges as an essential research imperative. In this paper, we propose CoGenesis, a collaborative generation framework integrating large (hosted on cloud infrastructure) and small models (deployed on local devices) to address privacy concerns logically. Initially, we design a pipeline to create personalized writing instruction datasets enriched with extensive context details as the testbed of this research issue. Subsequently, we introduce two variants of CoGenesis based on sketch and logits respectively. Our experimental findings, based on our synthesized dataset and two additional open-source datasets, indicate that: 1) Large-scale models perform well when provided with user context but struggle in the absence of such context. 2) While specialized smaller models fine-tuned on the synthetic dataset show promise, they still lag behind their larger counterparts. 3) Our CoGenesis framework, utilizing mixed-scale models, showcases competitive performance, providing a feasible solution to privacy issues.

1 Introduction

Large Language Models (LLMs)¹ have demonstrated significant potential in advancing artificial intelligence, exhibiting exceptional ability in instruction following and achieving superior performance in various tasks such as writing, coding,

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¹This paper defines large LMs (LLMs) as both closed and open-source models, designed for universal application and advanced performance, and intended for cloud deployment. Conversely, small LMs (SLMs) refer to models tailored for specific tasks and deployed on local devices.

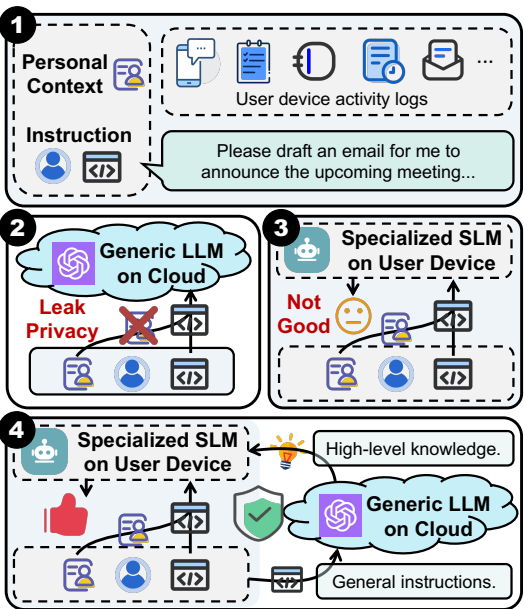


Figure 1: ① Context-aware instruction following example. ② LLMs excel with context but risk privacy. ③ Specialized and smaller LMs (SLMs) on device are privacy-friendly but underperform. ④ Collaborating LLMs and SLMs enhances privacy and performance.

and other text-based activities (Achiam et al., 2023; Bubeck et al., 2023; Touvron et al., 2023a,b). LLM-based AI assistants play a crucial role in executing instructions, aiding in writing tasks, and accelerating work processes, thereby fostering content innovation (Zhang et al., 2023a; Haase and Hanel, 2023). LLMs often require extensive context information for generating more personalized and effective content, owing to their in-context learning abilities (Brown et al., 2020). Retrieval Augmented Generation (RAG) (Gao et al., 2023) has proven beneficial in incorporating additional context to enhance the informativeness and personalization of LLM outputs. However, current instruction-following tasks often fail to consider rich user context in their design (Wang et al., 2023c; Xu et al., 2023b; Ding et al., 2023). Incorporating personal

CoGenesis: 一个用于协作大语言模型和小语言模型以安全上下文感知地遵循指令的框架

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摘要

随着语言模型 (LMs) 的发展, 它们接触到私人数据的情况越来越不可避免, 而且将它们 (尤其是较小的模型) 部署在个人设备上, 如 PC 和智能手机, 已成为一种普遍趋势。在充满用户信息的场景中, 使模型既能保护用户隐私, 又能高效执行命令, 成为一项重要的研究课题。在本文中, 我们提出了 CoGenesis, 一种整合大模型 (托管在云基础设施上) 和小模型 (部署在本地设备上) 的协作生成框架, 以合理地解决隐私问题。最初, 我们设计了一个管道, 用于创建个性化写作指令数据集, 并丰富了大量上下文细节, 作为这一研究问题的测试平台。随后, 我们基于草图和 logits 分别引入了两种 CoGenesis 变体。我们的实验结果基于我们合成的数据集和另外两个开源数据集表明:

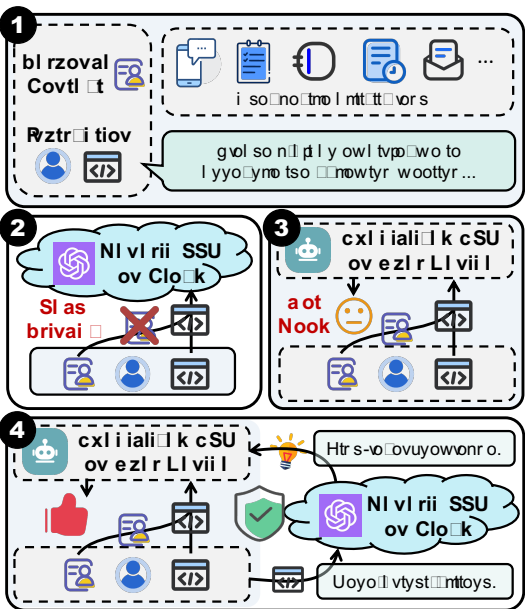


图 1: ① 上下文感知指令遵循示例。② 大型语言模型 (LLM) 在有上下文时表现出色, 但存在隐私风险。③ 专用的小型语言模型 (SLM) 在本地设备上运行对隐私友好, 但表现较差。④ LLM 与 SLM 的协作可提升隐私性和性能。

以及其他基于文本的活动 (Achiam 等, 2023; Bubeck 等, 2023; Touvron 等, 2023a,b)。基于大语言模型 (LLM) 的人工智能助手在执行指令、协助写作任务以及加速工作流程中起着关键作用, 从而促进内容创新 (Zhang 等, 2023a; Haase 和 Hanel, 2023)。LLM 通常需要大量上下文信息, 以生成更个性化和有效的内容, 这归功于其上下文学习能力 (Brown 等, 2020)。检索增强生成 (RAG) (Gao 等, 2023) 已被证明在引入额外上下文以提升 LLM 输出的信息量和个性化方面具有益处。然而, 目前的指令执行任务在设计中往往未能考虑丰富的用户上下文 (Wang 等, 2023c; Xu 等, 2023b; Ding 等, 2023)。整合个人

1 引言

大型语言模型 (LLMs) 在推动人工智能方面显示出显著潜力, 表现出卓越的指令遵循能力, 并在写作、编码等各类任务中取得优异表现,

通讯作者 本文将大型语言模型 (LLMs) 定义为既包括封闭源又包括开源模型, 旨在实现通用应用和高级性能, 并用于云端部署。相反, 小型语言模型 (SLMs) 指为特定任务定制并部署在本地设备上的模型。

experiences and activity logs could significantly augment the effectiveness of these instructions, called context-aware instruction following.

Despite their advancements, the most sophisticated LLMs, including GPT-4, Claude, and Gemini, are primarily commercialized and deployed on cloud services. This API-based deployment ensures the privacy of the LLMs but potentially compromises user privacy (Xiao et al., 2023; Zhang et al., 2023c). Although more powerful LLMs are being open-sourced, like Llama-2 (Touvron et al., 2023b), Qwen (Bai et al., 2023), and Mistral (Jiang et al., 2024), their stable deployment on local devices with limited resource remains challenging. Recent advancements in smaller LMs (SLMs) equipped with billions of parameters now enable their deployment on consumer desktops and smartphones, achieving satisfactory performance (Bai et al., 2023; Zhang et al., 2024; Singer et al., 2024; Hu et al., 2024). The balance between performance and privacy in LLMs and SLMs raises three critical questions: (1) *How effectively can LLMs operate without stringent user privacy contexts?* (2) *To what extent can specialized models, boasting billions of parameters, excel in context-aware instruction following?* (3) *Is it possible to navigate the trade-off between performance and privacy through collaborations between large and small models?*

As indicated in Figure 1, considering the following scenario in ④: smaller, personalized LMs are deployed on user devices with limited resources. These SLMs can access private data and activity logs on the devices while processing instructions. In contrast, the more advanced general LLMs operate on cloud services and receive only general instructions. In this setup, the LLMs provide high-level knowledge like deeper planning, superior outlines, and even “dark knowledge”. Meanwhile, the SLMs utilize the context information and knowledge provided by the LLMs to collaboratively generate personalized content. Current privacy protection methods for API-based services are limited (Cummings et al., 2023); they are either capable of handling only simple classification tasks with text sanitization (Kan et al., 2023; Chen et al., 2023b) or require encryption or noise addition (Zhou et al., 2022; Wu et al., 2023), which still pose a risk of data leakage. In contrast, the collaborative generation approach involving SLMs and LLMs can logically prevent privacy breaches without the need to upload context information.

Overall, we highlight our contributions as follows: (1) We introduce the context-aware instruction-following task that incorporates extensive user privacy context information. To support this, we design a four-step data construction process and synthesize a modest amount of instructional data for experimental validation. (2) We investigate the performance of LLMs and SLMs on this task. Our findings indicate that SLMs, when provided with context, can outperform LLMs lacking context but lag behind the performance of LLMs equipped with context. (3) For context-aware instruction generation, we present the CoGenesis framework. CoGenesis comprises sketch-based and logit-based variants to facilitate collaboration between large and small language models. This approach not only safeguards context privacy but also ensures performance gains.

2 Context-aware Instruction Following

2.1 Task Definition

Current instruction formats either consist of standalone instructions or instructions accompanied by additional inputs (Wang et al., 2023c). While these instructions typically cover generic tasks such as writing, searching, and coding, the inputs often contain specific task information, such as tables and coding problems. We classify the context-aware instruction-following task within the domain of controllable conditional text generation (Zhang et al., 2023b), enriching standard instructions with additional personal context. The task $\mathcal{T}$, focused on generating a response r relevant to user-specific data marked by privacy and personal style, is directed by instructions t and contextual information $\mathcal{C}$. It underscores the significance of contextual integration for enhancing output personalization and relevance, formally expressed as:

$$\mathcal{T}(t, \mathcal{C}) : r = g(t, \mathcal{C}; \theta) \quad (1)$$

This generative model g with parameters θ aims to adaptively respond to user instructions within the nuanced context of individual data attributes.

2.2 Data Construction

As illustrated on the left side of Figure 2, we delineate a novel four-step pipeline for crafting context-aware instructions aimed at generating personalized and creative text with AI assistants for diverse user groups. Our methodology begins with the

经验和活动日志可以显著增强这些指令的有效性，这被称为情境感知指令遵循。

尽管取得了进展，最复杂的 LLM，包括 GPT-4、Claude 和 Gemini，主要还是商业化并部署在云服务上。这种基于 API 的部署确保了 LLM 的隐私，但可能会损害用户隐私（Xiao 等 Zhang 等）。虽然像 Llama-2（Touvron 等）、Qwen（Bai 等）和 Mistral（Jiang 等）这样更强大的 LLM 正在开源，但在有限资源的本地设备上稳定部署仍然具有挑战性。最近在配备了数十亿参数的小型 LM（SLM）方面的进展，使其能够部署在消费者桌面和智能手机上，并实现令人满意的性能（Bai 等；</p>
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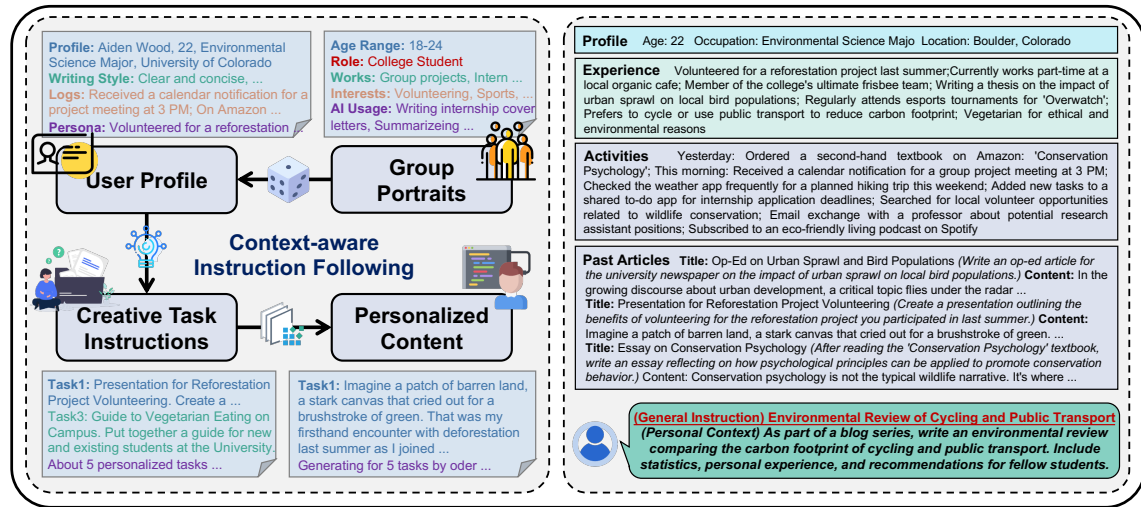


Figure 2: This illustration demonstrates construction process and example of context-aware instructions.

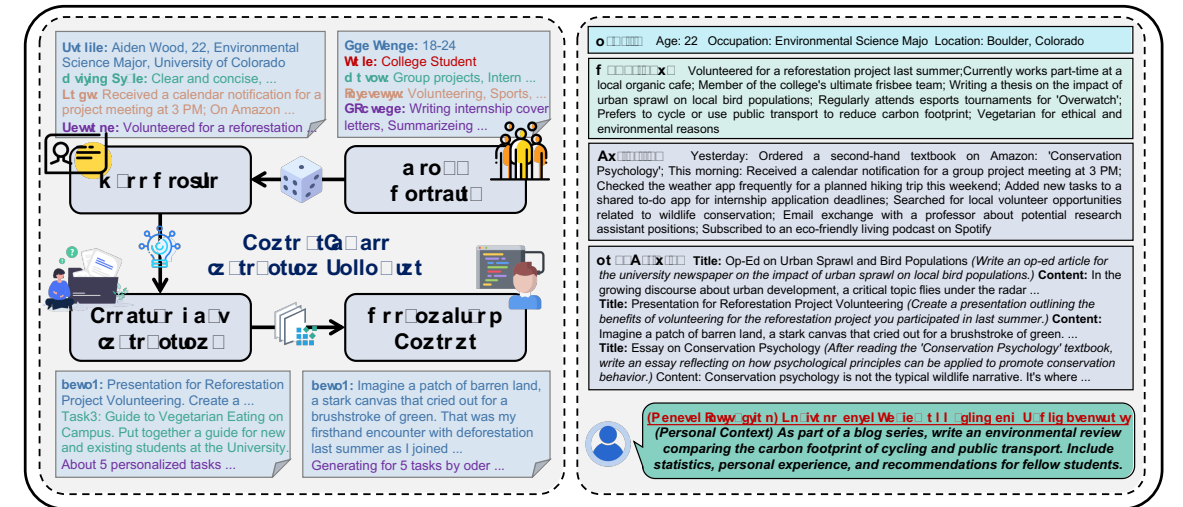


图 2: 此插图展示了构建过程和上下文感知指令的示例。

creation of detailed user group portraits, capturing demographics, professional backgrounds, and interests to identify specific AI application scenarios. Individual user profiles are then elaborated, incorporating unique writing styles, fictional personal details, and smart device usage to construct nuanced characters for AI writing tasks. These profiles inform the design of writing tasks that resonate with each character's lifestyle and digital interactions, ensuring task realism and relevance. Finally, we generate personalized texts that reflect the characters' professional and personal narratives with stylistic accuracy, demonstrating our approach's efficacy in producing coherent, context-specific content for AI-facilitated text generation.

An example is illustrated on the right side of Figure 2, where the user instruction comprises a general section and a personal section. The latter, in conjunction with the provided profile, experience, activities, and previous articles, constitutes the privacy-sensitive context information.

3 Collaborative Generation Framework

3.1 Overview of CoGenesis

We present the CoGenesis framework that capitalizes on the strengths of two differently scaled models: a LLM with parameters θ_l and a SLM with parameters θ_s . This framework is centered around the fusion strategy, denoted as $f(\cdot)$, which intelligently combines the outputs from both models. Specifically, θ_l generates replies solely based on the general instruction t , while θ_s considers both user instruction t and additional personal context C for its output generation. The fusion strategy

$f(\cdot)$ aims to synergistically blend the outputs of $\theta_l(r|t)$ and $\theta_s(r|t, C)$. Intuitively, the combined performance is expected to not only surpass that of the individual models but also closely match the performance of θ_l had it processed both r and C .

In our collaborative framework, sketches (or outlines) of content and next token logits from LLMs are considered forms of high-level knowledge. The two approaches to the function $f(\cdot)$ are identified as sketch-based and logit-based, respectively. The sketch-based approach is model-agnostic, whereas the logit-based method requires LLMs and SLMs to share the same tokenizer in our present configuration. In the following sections, we will detail these two implementations of $f(\cdot)$ sequentially.

3.2 Sketch-based CoGenesis

Recognizing the strengths of LLMs in planning and SLMs in crafting contextualized responses, we introduce a "sketch-then-fill" approach to synergize their capabilities for personalized content generation. As depicted on the left side of Figure 3, this approach consists of two crucial steps:

Step1: Sketch Generation by LLMs. Given the substantial cost and complexity, especially with API-dependent LLMs, we simplify the process by directly prompting LLMs with a general instruction t . The sketch r_{sketch} of content is derived through text decoding from the LLMs using a sampling strategy, succinctly represented as:

$$r_{\text{sketch}} = \text{Decoding}_{\text{LLM}}(t; \theta_l) \quad (2)$$

Step2: Content Personalization by SLMs. After acquiring the sketch r_{sketch} , the SLM utilizes it,

创建详细的用户群画像，捕捉人口统计、职业背景和兴趣，以识别特定的人工智能应用场景。然后详细描述个人用户档案，融入独特的写作风格、虚构的个人信息以及智能设备使用情况，以构建用于人工智能写作任务的细致人物。这些档案为设计与每个人物生活方式和数字互动相契合的写作任务提供依据，确保任务的现实性和相关性。最后，我们生成个性化文本，准确反映人物的职业和个人经历，展示我们的方法在为人工智能辅助文本生成生成连贯、特定情境内容方面的有效性。

示例在图 2 的右侧进行了说明，其中用户指令包括通用部分和个人部分。后者结合所提供的个人资料、经验、活动和先前文章，构成了隐私敏感的上下文信息。

3 协作生成框架

3.1 CoGenesis 概述

我们提出了 CoGenesis 框架，该框架利用两种不同规模模型的优势：一个参数为 θ_l 的大语言模型（LLM）和一个参数为 θ_s 的小语言模型（SLM）。该框架以融合策略为核心，记作 $f(\cdot)$ ，可以智能地结合两种模型的输出。具体来说， θ_l 仅根据通用指令 t 生成回复，而 θ_s 在生成输出时同时考虑用户指令 t 和额外的个人上下文 C 。融合策略

$f(\cdot)$ 旨在协同融合 θ_l 和 θ_s 的输出。从直观上看，组合后的性能预计不仅会超越各个模型的性能，还会接近 θ_l 如果它处理了 t 和 C 的性能。

在我们的协作框架中，内容的草图（或轮廓）以及来自大型语言模型（LLM）的下一个标记的 logits 被视为高层次知识的形式。函数 $f(\cdot)$ 的两种方法分别被识别为基于草图的方法和基于 logit 的方法。基于草图的方法与模型无关，而基于 logit 的方法在我们当前的配置中要求 LLM 和 SLM 使用相同的分词器。在接下来的章节中，我们将依次详细介绍函数 $f(\cdot)$ 的这两种实现。

3.2 基于草图的共生

认识到大型语言模型（LLMs）在规划方面的优势以及小型语言模型（SLMs）在制作情境化回应方面的能力，我们引入了一种“先草拟-再LL”的方法，以协同它们在个性化内容生成中的能力。如图 2 左侧所示，该方法包括两个关键步骤：

步骤 1: 通过大语言模型生成草图。鉴于成本高且复杂，尤其是依赖 API 的大语言模型，我们通过直接向大语言模型提供通用指令 t 来简化这一过程。内容的草图 r_{sketch} 是通过使用采样策略从大语言模型进行文本解码而得出的，可以简明地表示为：

$$r_{\text{sketch}} = \text{Decoding}_{\text{LLM}}(t; \theta_l) \quad (2)$$

步骤 2: 通过 SLM 进行内容个性化。在获取草图之后，SLM 会利用它，

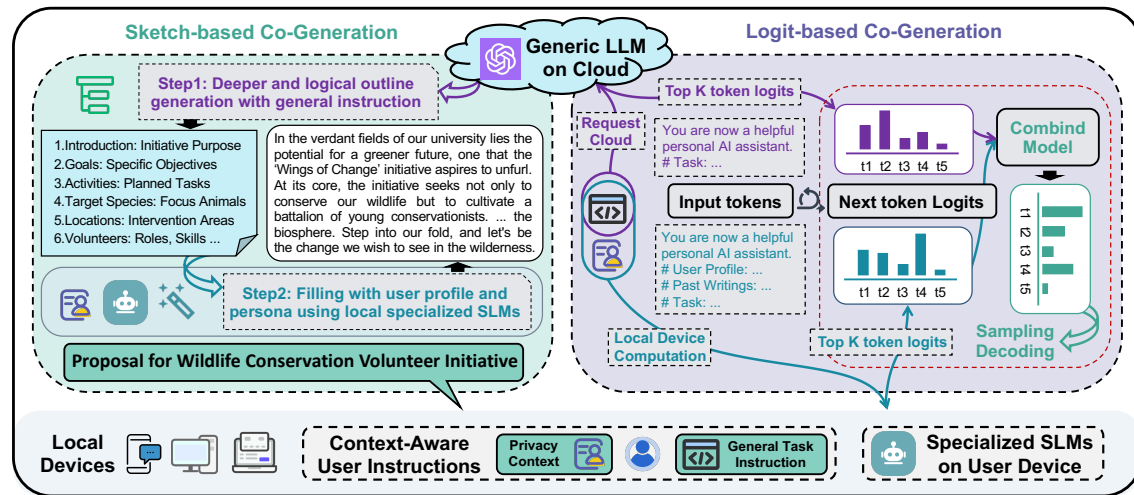


Figure 3: This figure demonstrates two collaborative generation variants of CoGenesis framework.

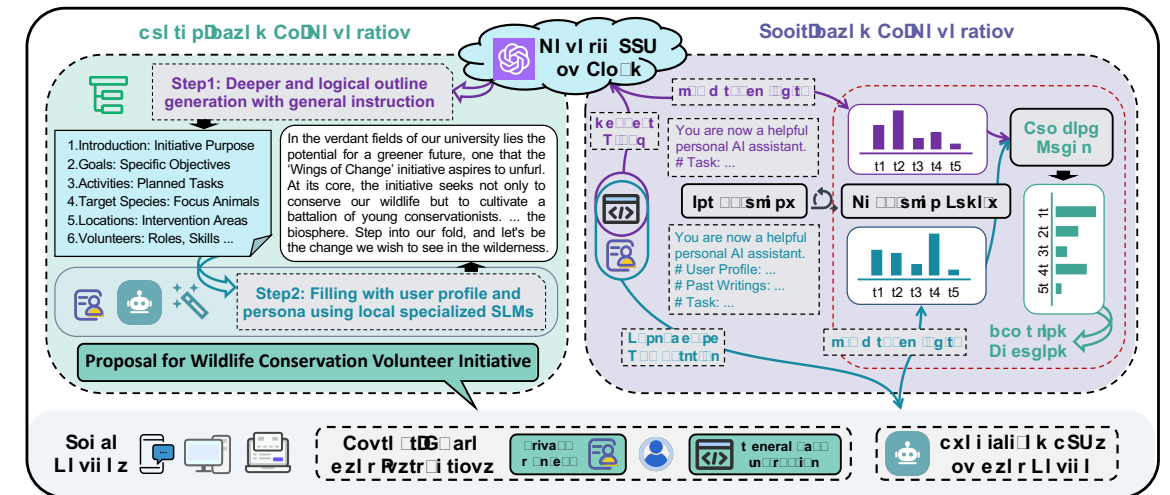


图 3: 此图展示了 CoGenesis 框架的两种协作生成变体。

along with the initial instruction t and personal context $\mathcal{C}$, to tailor personalized content r . This phase focuses on fine-tuning the SLM's parameters θ_s to optimize content relevance and personalization:

$$\hat{r} = \arg \max_r P(r|t, \mathcal{C}, r_{\text{sketch}}; \theta_s) \quad (3)$$

In this formula, r denotes the final, customized content, and P is the likelihood of generating r given instruction t , context $\mathcal{C}$, and sketch r_{sketch} with parameters θ_s . This approach delineates the use of LLMs for foundational text sketching based on user prompts, without necessitating parameter adjustments, and SLMs for further content refinement. This ensures the final output aligns with user specifications and their interaction history, highlighting the distinct yet complementary roles of LLMs and SLMs in personalized content creation.

3.3 Logit-based CoGenesis

The logits produced in the final layer of language models encapsulate a wealth of information, reflecting the models' internal dark knowledge. Previous efforts, such as contrastive decoding (Li et al., 2022) and emulator tuning (Mitchell et al., 2023), have explored the synergistic use of logits from both LLMs and SLMs to diminish hallucinations (Sennrich et al., 2023), augment reasoning capabilities (O'Brien and Lewis, 2023), and streamline the fine-tuning process of LLMs (Liu et al., 2024). Motivated by these works, our logit-based strategy involves integrating the logits of LLMs and SLMs under different inputs, ensuring collaborative determination of the subsequent token. A notable aspect of our method is the differential context exposure for the models: SLMs access the

full privacy context, while LLMs are provided with only broad instructions, as shown in Figure 3.

Defining the response sequence up to the k th token as $r_{<k}$, and denoting the k th token probabilities over vocabulary generated by LLMs and SLMs as p_k^l and p_k^s respectively, we leverage a lightweight combined model, denoted as CombModel, with parameters θ_c , to derive fusion weights w for final combined probabilities p_k^c . The computation of the logit-based method proceeds as follows:

$$p_k^s = \theta_s(r_{<k}, t, \mathcal{C}), p_k^l = \theta_l(r_{<k}, t) \quad (4)$$

$$w = \text{CombModel}(p_k^l, p_k^s) \quad (5)$$

$$p_k^c = w \cdot p_k^s + (1 - w) \cdot p_k^l \quad (6)$$

4 Experiments

4.1 Datasets

Synthetic Dataset. Following the construction process outlined in § 2.2, we synthesize a context-aware instruction-following dataset with GPT-4. Due to cost considerations, we construct a dataset representing thousands of fictitious users, based on hundreds of group portraits. After conducting quality and format filtering, we obtained a total of 1,500 users for training and validation. Additionally, we selected approximately 200 users from diverse group portraits to serve as the test set.

Open-source Datasets. In addition to our synthesized context-aware instruction datasets, we also utilize publicly accessible, personalized context writing datasets, although they are limited to specific tasks in domains such as email and academic

结合初始指令 和个人上下文 , 定制个性化内容 。此阶段侧重于微调 SLM 的参数以优化内容相关性和个性化 :

$$\hat{r} = \arg \max_r P(r|t, \mathcal{C}, r_{\text{sketch}}; \theta_s) \quad (3)$$

在这个公式中, 表示最终的、定制化的内容, 是给定指令 、上下文 以及草稿 的情况下生成的概率 , 参数为 。这种方法阐明了在不需要调整参数的情况下, 如何利用大型语言模型

(LLM) 进行基于用户提示的基础文本草稿创作, 以及使用小型语言模型 (SLM) 进行进一步内容精炼。这确保最终输出与用户的具体要求及其交互历史相一致, 并凸显出 LLM 和 SLM 在个性化内容创作中既不同又互补的作用。

3.3 基于 Logit 的 CoGenesis

语言模型最后一层产生的 logits 蕴含丰富信息, 反映了模型的内部黑暗知识。以往的研究, 如对比解码 (Li 等,) 和模拟器调优 (Mitchell 等,), 已经探索了将 LLM 和 SLM 的 logits 协同使用, 以减少幻觉 (Sennrich 等,)、增强推理能力 (O'Brien 和 Lewis,) 以及简化 LLM 的微调过程 (Liu 等,)。受到这些工作的启发, 我们的基于 logits 的策略涉及在不同输入下整合 LLM 与 SLM 的 logits, 确保协作决定下一个 token。我们方法的一个显著特点是模型的上下文暴露不同: SLM 可以访问 th

完整的隐私上下文, 而 LLM 只提供了宽泛的指令, 如图 所示。

将响应序列定义到第 个标记为 , 并将第 个标记在词汇表上的概率由 LLM 和 SLM 生成分别记为 和 , 我们利用一个轻量级的组合模型, 记为 CombModel, 参数为 , 来推导最终组合概率 的融合权重 。基于 logit 的方法计算过程如下:

$$p_k^s = \theta_s(r_{<k}, t, \mathcal{C}), p_k^l = \theta_l(r_{<k}, t) \quad (4)$$

$$w = \text{CombModel}(p_k^l, p_k^s) \quad (5)$$

$$p_k^c = w \cdot p_k^s + (1 - w) \cdot p_k^l \quad (6)$$

4 个实验

4.1 数据集

合成数据集。按照 中概述的构建过程, 我们使用 GPT-4 合成了一个上下文感知的指令跟随数据集。由于成本考虑, 我们基于数百张群体画像构建了一个代表数千名虚构用户的数据集。在进行质量和格式筛选后, 我们最终获得了 1,500 名用户用于训练和验证。此外, 我们从多样化的群体画像中选择了大约 200 名用户作为测试集。

开源数据集。除了我们的合成上下文感知指令数据集外, 我们还使用公开可访问的个性化上下文写作数据集, 尽管它们仅限于电子邮件和学术等领域的特定任务。

papers. Specifically, we employ the processed Avocado Research Email and Citation Network Papers datasets in LaMP (Salemi et al., 2023). Furthermore, we refine these datasets to facilitate the generation of email bodies and paper abstracts, considering previous emails and papers from the same users as contextual information.

Further details about the synthesized and processed datasets can be found in Appendix A.

4.2 Baselines

Settings. We primarily evaluate four configurations in our experiments: **1) LLM with context.** We engage LLMs with additional context to facilitate personalized generation. It’s an *upper bound* that may compromise context privacy. **2) LLM w/o context.** Given the importance of privacy in contextual data, it is advisable to limit requests to cloud-based LLMs to general instructions only. This setting serves as a *lower bound* that preserves user privacy. **3) SLM with context.** This setting establishes the baseline for privacy-protected, on-device personalized generation. **4) SLM + LLM with context.** This is our proposed collaborative generation between large- and small-scale models. Within this framework, we evaluate sketch-based and logit-based methods.

Models. Our selection of LLMs encompasses both commercial API-based and open-source models. For the commercial segment, we concentrate on GPT-3.5-turbo and GPT-4-turbo. In the realm of open-source, we opt for the largest models within the Llama-2 (Touvron et al., 2023b), Qwen (72B in versions 1 and 1.5 ²) (Bai et al., 2023), and Mistral (Jiang et al., 2024) series. Regarding SLMs, we prioritize the most recently released models with 1~2 billions of parameters. This includes TinyLlama (Zhang et al., 2024), Qwen (1.8B in versions 1 and 1.5), StableLM ³, and H2O-Danube (Singer et al., 2024). For both LLMs and SLMs, our focus is on the chat versions, employing the default template for each model for consistency.

Additionally, zero-shot LLMs are used in both *with context* and *w/o context* settings. For *SLM with context*, we include both zero-shot and fine-tuned models, whereas only fine-tuned SLMs are utilized in *SLM + LLM with context*. Further information on prompts for LLMs and fine-tuning details for SLMs is presented in Appendix B.

4.3 Evaluation Metrics

For instruction-following evaluation, LLM-based evaluators like GPT-4 have shown high consistency and effectiveness (Chang et al., 2023), particularly in the generation of personalized content, outperforming human evaluators in consistency (Wang et al., 2023b). Therefore, we employ GPT-4-turbo as a judge to assess generated content from multiple perspectives. This evaluation encompasses several criteria: personalization, alignment with the user profile, helpfulness, relevance, depth, creativity, and the level of detail, collectively contributing to the overall score (Ovl.). To further analyze the performance, we introduce two distinct prompts: one incorporating user context (Ovl.(w)) and the other excluding it (Ovl.(w/o)), enabling a comparative assessment of LLMs in both personalized content generation and broad instruction adherence. Additionally, the personalized scores of the responses are assessed independently, indicated as **Per**.

4.4 Main Results

As illustrated in Table 1, we analyze the experimental results considering the following aspects:

Results on LLMs. In the *with context* setting, GPT-4-turbo outperformed all other models across all metrics and datasets, followed by Qwen-Chat(v1.5) with the second-best performance. Similar outcomes were observed in the *w/o context* setting. However, a comparison between the two settings reveals that all LLMs exhibit diminished performance in terms of personalization and overall scores in the absence of context information, underscoring the value of context. Interestingly, while context greatly influences the scores in *with context* setting, it has a negligible effect on scores in *w/o context* setting, reflecting advanced capability of LLMs in adhering to general instructions. Moreover, when comparing our synthesized context-aware dataset and the email dataset to the paper abstract dataset, the latter demonstrates limited personalization factors resulting in high performance in *w/o context* setting.

Results on SLMs. Zero-shot SLMs exhibit highly varied performances, influenced by pre-training and supervised fine-tuning factors. Notably, StableLM-Zephyr achieves the highest performance, with H2O-Danube-Chat and Qwen-Chat(v1.5) closely competing for second place. Interestingly, with the advantage of context, zero-shot SLMs can outperform LLMs in scenarios

论文。具体来说，我们使用 LaMP 中处理过的鳄梨研究电子邮件和引用网络论文数据集 (Salemi 等,)。此外，我们进一步优化这些数据集，以便生成电子邮件正文和论文摘要，同时将同一用户之前的电子邮件和论文作为上下文信息考虑在内。

关于合成和处理过的数据集的更多详细信息可以在附录中找到。

4.2 基线

设置。我们在实验中主要评估四种配置：带上下文的 LLM。我们为 LLM 提供额外的上下文，以便生成个性化内容。这是一个上限，可能会影响上下文隐私。不带上下文的 LLM。鉴于上下文数据隐私的重要性，建议将对基于云的 LLM 的请求限制为一般指令。此设置作为下限，能够保护用户隐私。带上下文的 SLM。该设置建立了保护隐私的本地个性化生成基线。带上下文的 LLM。这是我们提出的大型模型与小型模型协作生成的方法。在此框架下，我们评估基于草图和基于 logit 的方法。

模型。我们的 LLM 选择涵盖了商业 API 型和开源模型。在商业部分，我们主要关注 GPT-3.5-turbo 和 GPT-4-turbo。在开源领域，我们选择了 Llama-2 (Touvron 等,)、Qwen (版本 1 和 1.5 的 72B) (Bai 等,) 以及 Mistral (Jiang 等,) 系列中最大的模型。关于 SLM，我们优先选择最近发布的拥有数十亿参数的模型。这包括 TinyLlama (Zhang 等,)、Qwen (版本 1 和 1.5 的 1.8B)、StableLM 以及 H2O-Danube (Singer 等,)。对于 LLM 和 SLM，我们的重点是聊天版本，使用每个模型的默认模板进行协作

此外，零样本大语言模型 (LLM) 在有上下文的和无上下文的设置中都被使用。对于有上下文的序列语言模型 (SLM)，我们同时包含零样本和微调模型，而在 LLM 有上下文的 SLM 中仅使用微调的 SLM。关于 LLM 的提示和 SLM 的微调细节的更多信息，请参见附录。

²<https://github.com/QwenLM/Qwen1.5>
³<https://huggingface.co/stabilityai>

4.3 评估指标

在指令遵循评估中，基于大语言模型 (LLM) 的评估者如 GPT-4 已显示出高度的一致性和有效性 (Chang 等)，尤其是在个性化内容生成方面，其一致性优于人工评估者 (Wang 等)。因此，我们采用 GPT-4-turbo 作为评判者，从多个角度评估生成的内容。本次评估涵盖若干标准：个性化、与用户特征的匹配性、帮助性、相关性、深度、创造力以及细节水平，这些标准共同构成整体评分 ()。为了进一步分析性能，我们引入了两个不同的提示：一个包含用户上下文 ()，另一个不包含用户上下文 ()，从而能够对 LLM 在个性化内容生成和广义指令遵循方面进行比较评估。A

4.4 主要结果

如表 所示，我们从以下几个方面分析实验结果：

关于大型语言模型的结果。在有上下文设置中，GPT-4-turbo 在所有指标和数据集上均优于其他模型，其次是 Qwen-Chat(v1.5)，表现第二好。在无上下文设置中也观察到类似的结果。然而，将两种设置进行比较发现，在缺乏上下文信息的情况下，所有大型语言模型在个性化和整体评分方面的表现都会下降，这凸显了上下文的重要性。有趣的是，虽然上下文在有上下文设置中对评分有很大影响，但在无上下文设置中对评分的影响可忽略不计，这反映了大型语言模型在遵循一般指令方面的高级能力。此外，当将我们合成的上下文感知数据集和电子邮件数据集与论文摘要数据集进行比较时，后者表现出有限的个性化因素，从而在无上下文设置中表现出较高性能。

SLM 的结果。零样本 SLM 表现出高度变化的性能，受预训练和监督微调因素的影响。值得注意的是，StableLM-Zephyr 取得了最高性能，H2O-Danube-Chat 和 Qwen-Chat(v1.5) 紧随其后，竞争第二名。有趣的是，在上下文优势下，零样本 SLM 在某些场景中可以超过 LLM 的表现。

²<https://github.com/QwenLM/Qwen1.5>
³<https://huggingface.co/stabilityai>

Model	Params	Context-aware Instructions			Avocado Emails			Academic Paper Abstracts		
		Ovl.(w)	Per.	Ovl.(w/o)	Ovl.(w)	Per.	Ovl.(w/o)	Ovl.(w)	Per.	Ovl.(w/o)
zero-shot LLM with context (upper bound)										
L1 GPT-4-turbo	N/A	8.85	8.90	8.54	8.31	7.71	8.05	8.64	8.47	8.68
L2 GPT-3.5-turbo	N/A	8.30	8.33	7.58	7.70	7.45	7.57	7.97	7.88	8.29
L3 Llama-2-Chat	70B	7.78	7.98	7.96	7.00	7.18	8.05	7.48	7.12	8.32
L4 Qwen-Chat(v1)	72B	8.38	8.38	8.14	7.62	7.10	7.70	7.90	7.62	8.24
L5 Qwen-Chat(v1.5)	72B	8.70	8.67	8.26	8.20	7.69	7.80	8.52	8.16	8.60
L6 Mixtral-8x7b	47B	8.12	8.22	7.96	7.35	6.88	6.92	7.92	7.62	8.08
zero-shot LLM w/o context (lower bound)										
L1 GPT-4-turbo	N/A	6.10	6.46	8.75	5.10	3.79	8.58	8.42	7.94	8.73
L2 GPT-3.5-turbo	N/A	4.34	3.76	7.47	3.72	3.03	7.61	3.37	3.59	6.04
L3 Llama-2-Chat	70B	4.74	4.60	8.12	3.38	3.52	8.04	6.74	6.18	7.90
L4 Qwen-Chat(v1)	72B	3.70	3.38	7.72	3.28	2.38	7.50	6.42	5.54	7.90
L5 Qwen-Chat(v1.5)	72B	5.86	5.98	8.52	5.83	4.40	8.54	7.92	7.16	8.28
L6 Mixtral-8x7b	47B	5.32	5.08	8.14	3.38	3.52	8.04	6.74	6.18	7.90
zero-shot SLM with context										
S1 StableLM-Zephyr	1.6B	6.88	6.82	6.68	6.03	5.49	6.51	7.32	6.94	8.14
S2 H2O-danube-chat	1.8B	6.56	6.94	6.60	4.77	5.00	6.57	5.98	5.54	6.94
S3 TinyLlama-Chat	1.1B	1.72	1.84	2.14	4.00	3.54	5.40	1.96	1.88	4.24
S4 Qwen-Chat(v1)	1.8B	5.78	5.50	6.00	4.91	4.54	6.49	4.46	5.04	6.74
S5 Qwen-Chat(v1.5)	1.8B	6.86	6.86	7.20	5.51	4.89	6.14	6.42	6.14	7.78
finetuned SLM (+ LLM) with context										
S1 StableLM-Zephyr + L1 sketch	1.6B mixed	8.30 8.48 ^{↑0.18}	8.56 8.56 ^{↑0.00}	7.73 7.98 ^{↑0.25}	7.58 7.68 ^{↑0.10}	6.70 6.62 ^{↓0.08}	7.20 7.48 ^{↑0.28}	7.96 8.28 ^{↑0.32}	7.64 7.48 ^{↓0.16}	8.18 8.38 ^{↑0.20}
S2 H2O-danube-chat + L1 sketch	1.8B mixed	7.64 7.84 ^{↑0.20}	7.58 7.78 ^{↑0.20}	7.00 7.14 ^{↑0.14}	6.50 7.14 ^{↑0.64}	6.16 6.72 ^{↑0.56}	6.34 7.52 ^{↑1.18}	7.70 8.10 ^{↑0.40}	7.30 7.28 ^{↓0.02}	8.06 8.18 ^{↑0.12}
S3 TinyLlama-Chat + L1 sketch + L3 logits	1.1B mixed mixed	7.42 7.66 ^{↑0.24} 7.76 ^{↑0.34}	7.66 7.14 ^{↓0.52} 7.74 ^{↑0.08}	6.78 6.82 ^{↑0.04} 7.06 ^{↑0.28}	6.12 6.58 ^{↑0.46} 6.06 ^{↓0.06}	5.92 6.02 ^{↑0.10} 6.16 ^{↑0.24}	6.20 6.60 ^{↑0.40} 6.94 ^{↑0.74}	7.66 7.72 ^{↑0.06} 8.14 ^{↑0.48}	7.32 7.36 ^{↑0.04} 7.34 ^{↑0.02}	8.18 8.10 ^{↓0.08} 8.04 ^{↓0.14}
S4 Qwen-Chat(v1) + L1 sketch + L4 logits	1.8B mixed mixed	7.44 7.80 ^{↑0.36} 8.12 ^{↑0.68}	7.76 7.82 ^{↑0.06} 8.20 ^{↑0.44}	7.02 7.64 ^{↑0.62} 7.86 ^{↑0.84}	7.00 7.18 ^{↑0.18} 7.48 ^{↑0.48}	6.71 6.44 ^{↓0.27} 6.44 ^{↓0.27}	7.06 7.28 ^{↑0.22} 7.46 ^{↑0.40}	7.84 8.02 ^{↑0.18} 7.92 ^{↑0.08}	7.36 7.70 ^{↑0.34} 7.16 ^{↓0.20}	8.18 8.34 ^{↑0.16} 8.30 ^{↑0.12}
S5 Qwen-Chat(v1.5) + L1 sketch + L5 logits	1.8B mixed mixed	8.08 8.18 ^{↑0.10} 8.28 ^{↑0.20}	8.12 7.98 ^{↓0.14} 8.22 ^{↑0.10}	7.40 7.62 ^{↑0.22} 7.80 ^{↑0.40}	6.34 6.54 ^{↑0.20} 6.66 ^{↑0.32}	5.56 5.84 ^{↑0.28} 5.70 ^{↑0.14}	6.54 6.74 ^{↑0.20} 7.12 ^{↑0.58}	7.68 8.10 ^{↑0.42} 8.14 ^{↑0.46}	7.30 7.24 ^{↓0.06} 7.28 ^{↓0.02}	8.18 8.32 ^{↑0.14} 8.40 ^{↑0.22}

Table 1: The table displays the performance of LLMs and SLMs in both *with context* and *w/o context* settings. We highlight the best result in **bold**, the second in underline and indicate variations in each SLM using [↑] and [↓].

lacking context. However, LLMs consistently outperform SLMs in overall scores without context, underscoring their superior capabilities in general instruction generation. After fine-tuning, SLMs surpass the performance of many LLMs with context, demonstrating the benefits of specialization. Yet, SLMs do not reach the performance levels of the most powerful LLMs, such as GPT-4-turbo and Qwen-Chat(v1.5). These results highlight the promise of collaboration between specialized SLMs and LLMs for achieving better personalized scores, deeper writing, and enhanced instruction generalization.

Results on Mixed-Scale Models Collaborations. In our exploration of mixed-scale model collaboration, we utilize the sketch-based method for all SLMs and the logit-based method exclusively within the Llama and Qwen model families. This comparison reveals that collaborations between mixed-scale models achieve results comparable to those of LLMs alone, while also safe-

guarding privacy. Collaborative efforts generally enhance the overall scores of SLMs, both in evaluations with and without user context. Nevertheless, incorporating sketch-based collaboration, in particular, might slightly detract from personalization scores due to the reliance on LLMs that inference without context. Between the logit-based and sketch-based approaches, the former proves more efficacious, contingent upon the SLMs and LLMs utilizing a common tokenizer. Overall, this collaborative strategy between mixed-scale models offers a promising avenue for balancing efficiency and privacy considerations, though it still necessitates further refinements to optimize performance.

4.5 Ablation Study

4.5.1 Sketch-based CoGenesis

Sketch vs. Full Content. Figure 4 contrasts the efficacy of employing merely the sketch versus the entire content provided by LLMs. The findings indicate that incorporating full content generally

Model	Params	Context-aware Instructions			Avocado Emails			Academic Paper Abstracts		
		Ovl.(w)	Per.	Ovl.(w/o)	Ovl.(w)	Per.	Ovl.(w/o)	Ovl.(w)	Per.	Ovl.(w/o)
zero-shot LLM with context (upper bound)										
L1 GPT-4-turbo	N/A	8.85	8.90	8.54	8.31	7.71	8.05	8.64	8.47	8.68
L2 GPT-3.5-turbo	N/A	8.30	8.33	7.58	7.70	7.45	7.57	7.97	7.88	8.29
L3 Llama-2-Chat	70B	7.78	7.98	7.96	7.00	7.18	8.05	7.48	7.12	8.32
L4 Qwen-Chat(v1)	72B	8.38	8.38	8.14	7.62	7.10	7.70	7.90	7.62	8.24
L5 Qwen-Chat(v1.5)	72B	8.70	8.67	8.26	8.20	7.69	7.80	8.52	8.16	8.60
L6 Mixtral-8x7b	47B	8.12	8.22	7.96	7.35	6.88	6.92	7.92	7.62	8.08
zero-shot LLM w/o context (lower bound)										
L1 GPT-4-turbo	N/A	6.10	6.46	8.75	5.10	3.79	8.58	8.42	7.94	8.73
L2 GPT-3.5-turbo	N/A	4.34	3.76	7.47	3.72	3.03	7.61	3.37	3.59	6.04
L3 Llama-2-Chat	70B	4.74	4.60	8.12	3.38	3.52	8.04	6.74	6.18	7.90
L4 Qwen-Chat(v1)	72B	3.70	3.38	7.72	3.28	2.38	7.50	6.42	5.54	7.90
L5 Qwen-Chat(v1.5)	72B	5.86	5.98	8.52	5.83	4.40	8.54	7.92	7.16	8.28
L6 Mixtral-8x7b	47B	5.32	5.08	8.14	3.38	3.52	8.04	6.74	6.18	7.90
zero-shot SLM with context										
S1 StableLM-Zephyr	1.6B	6.88	6.82	6.68	6.03	5.49	6.51	7.32	6.94	8.14
S2 H2O-danube-chat	1.8B	6.56	6.94	6.60	4.77	5.00	6.57	5.98	5.54	6.94
S3 TinyLlama-Chat	1.1B	1.72	1.84	2.14	4.00	3.54	5.40	1.96	1.88	4.24
S4 Qwen-Chat(v1)	1.8B	5.78	5.50	6.00	4.91	4.54	6.49	4.46	5.04	6.74
S5 Qwen-Chat(v1.5)	1.8B	6.86	6.86	7.20	5.51	4.89	6.14	6.42	6.14	7.78
netuned SLM (+ LLM) with context										
S1 StableLM-Zephyr	1.6B	8.30	8.56	7.73	7.58	6.70	7.20	7.96	7.64	8.18
+ L1 sketch	mixed	8.48 _{↑0.18}	8.56 _{↑0.00}	7.98 _{↑0.25}	7.68 _{↑0.10}	6.62 _{↓0.08}	7.48 _{↑0.28}	8.28 _{↑0.32}	7.48 _{↓0.16}	8.38 _{↑0.20}
S2 H2O-danube-chat	1.8B	7.64	7.58	7.00	6.50	6.16	6.34	7.70	7.30	8.06
+ L1 sketch	mixed	7.84 _{↑0.20}	7.78 _{↑0.20}	7.14 _{↑0.14}	7.14 _{↑0.64}	6.72 _{↑0.56}	7.52 _{↑1.18}	8.10 _{↑0.40}	7.28 _{↓0.02}	8.18 _{↑0.12}
S3 TinyLlama-Chat	1.1B	7.42	7.66	6.78	6.12	5.92	6.20	7.66	7.32	8.18
+ L1 sketch	mixed	7.66 _{↑0.24}	7.14 _{↓0.52}	6.82 _{↑0.04}	6.58 _{↑0.46}	6.02 _{↑0.10}	6.60 _{↑0.40}	7.72 _{↑0.06}	7.36 _{↑0.04}	8.10 _{↓0.08}
+ L3 logits	mixed	7.76 _{↑0.34}	7.74 _{↑0.08}	7.06 _{↑0.28}	6.06 _{↓0.06}	6.16 _{↑0.24}	6.94 _{↑0.74}	8.14 _{↑0.48}	7.34 _{↑0.02}	8.04 _{↓0.14}
S4 Qwen-Chat(v1)	1.8B	7.44	7.76	7.02	7.00	6.71	7.06	7.84	7.36	8.18
+ L1 sketch	mixed	7.80 _{↑0.36}	7.82 _{↑0.06}	7.64 _{↑0.62}	7.18 _{↑0.18}	6.44 _{↓0.27}	7.28 _{↑0.22}	8.02 _{↑0.18}	7.70 _{↑0.34}	8.34 _{↑0.16}
+ L4 logits	mixed	8.12 _{↑0.68}	8.20 _{↑0.44}	7.86 _{↑0.84}	7.48 _{↑0.48}	6.44 _{↓0.27}	7.46 _{↑0.40}	7.92 _{↑0.08}	7.16 _{↓0.20}	8.30 _{↑0.12}
S5 Qwen-Chat(v1.5)	1.8B	8.08	8.12	7.40	6.34	5.56	6.54	7.68	7.30	8.18
+ L1 sketch	mixed	8.18 _{↑0.10}	7.98 _{↓0.14}	7.62 _{↑0.22}	6.54 _{↑0.20}	5.84 _{↑0.28}	6.74 _{↑0.20}	8.10 _{↑0.42}	7.24 _{↓0.06}	8.32 _{↑0.14}
+ L5 logits	mixed	8.28 _{↑0.20}	8.22 _{↑0.10}	7.80 _{↑0.40}	6.66 _{↑0.32}	5.70 _{↑0.14}	7.12 _{↑0.58}	8.14 _{↑0.46}	7.28 _{↓0.02}	8.40 _{↑0.22}

表 1: 该表显示了 LLM 和 SLM 在有上下文和无上下文设置下的表现。我们将最佳结果**标粗**，第二名下划线，并使用[↑]和[↓]标注每个 SLM 的变动。

缺乏上下文。然而，LLM 在没有上下文的情况下整体得分始终优于 SLM，突显了其在通用指令生成方面的卓越能力。经过微调后，SLM 的表现超过了许多有上下文的 LLM，显示了专业化的优势。然而，SLM 的表现仍未达到最强大的 LLM（如 GPT-4-turbo 和 Qwen-Chat(v1.5)）的水平。这些结果凸显了专用 SLM 与 LLM 协作的潜力，以实现更好的个性化得分、更深入的写作以及增强的指令泛化能力。

混合尺度模型合作的结果。在我们对混合尺度模型合作的探索中，我们对所有 SLM 使用基于草图的方法，而在 Llama 和 Qwen 模型家族中则仅使用基于逻辑的方法。此比较表明，混合尺度模型之间的合作所取得的结果可与单独的 LLM 相媲美，同时也更加安全。

保护隐私。协作努力通常会提升 SLM 的总体评分，无论是在有用户上下文还是无用户上下文的评估中。然而，特别是融入基于草图的协作，可能会略微降低个性化评分，因为依赖于在没有上下文的情况下进行推理的 LLM。在基于 logit 和基于草图的方法之间，前者证明更为有效，这取决于 SLM 和 LLM 是否使用相同的分词器。总体而言，这种混合规模模型之间的协作策略为平衡效率和隐私考虑提供了一个有前景的途径，尽管它仍然需要进一步的优化以提高性能。

4.5 消融研究

4.5.1 基于草图的共生成

草图与完整内容。图 对比了仅使用草图与使用 LLM 提供的完整内容的效果。研究结果表明，通常加入完整内容更有利

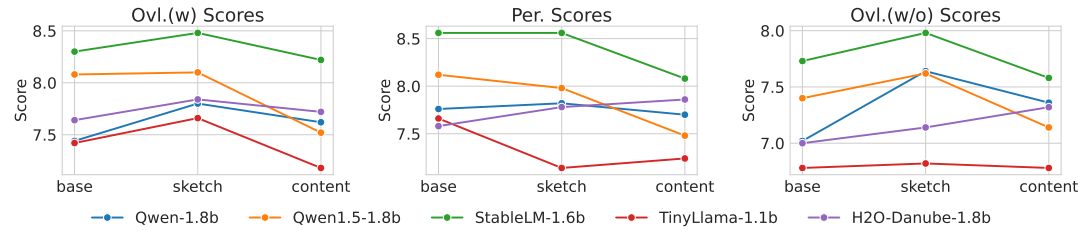


Figure 4: Comparative Performance of SLMs Utilizing Sketch versus Full Content.

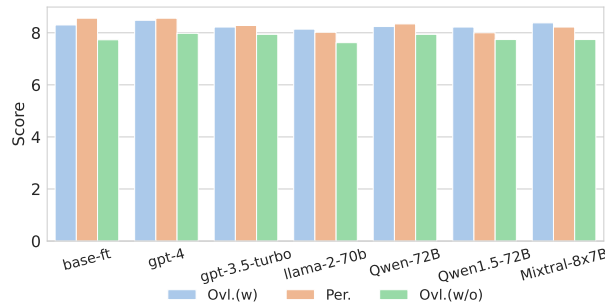


Figure 5: Performance of StableLM Using Sketches Generated by Various Models.

detracts from model performance, in contrast to utilizing no content or only the sketch. This discrepancy can be attributed to the potential overload of redundant information. As explored in (Weston and Sukhbaatar, 2023), an excess of content does not invariably enhance performance and risks the inclusion of extraneous details.

Generalization Capabilities of Sketch. For small models fine-tuned with sketches generated by GPT-4, we explore their generalization potential by utilizing sketches from a variety of LLMs during testing. Specifically, we focus on StableLM-Zephyr for detailed ablation analysis. Figure 5 demonstrates that employing sketches from alternative LLMs marginally affects the overall and personalized scores negatively but enhances the overall score in scenarios lacking context, relative to GPT-4. This suggests that sketches generated by different models vary and exhibit limited generalization capabilities.

4.5.2 Logit-based CoGenesis

Logits Fusing Strategy. We implement a learnable model designed for merging logits from context-free LLMs and context-inclusive SLMs in § 3.3. Additionally, we explore straightforward max and mean pooling strategies, acknowledged as robust baselines in (Ormazabal et al., 2023). According to Table 2, both max and mean pooling methods enhance the performance of SLMs, with max pooling

Metric	OvL(w)	Per.	OvL(w/o)
Models			
Qwen-72B-Chat (<i>with context</i>)	8.38	8.38	8.14
Qwen-72B-Chat (<i>w/o context</i>)	3.70	3.38	7.72
Qwen-1.8B-Chat (<i>finetuned</i>)	7.44	7.76	7.02
Mean Pooling Fusing	7.76	7.84	7.42
Max Pooling Fusing	7.90	7.94	7.52
Learnable Weights Fusing	8.12	8.20	7.86

Table 2: Comparison of Mean and Max Fusion Strategies Against our Learnable Fusion Model.

Model and Setting	Win/Tie/Lose (%)	BLEU	ROUGE-L
SLM finetuned	-/50/-	2.07	13.95
LLM w/ context	38/2/10	2.61	14.66
LLM w/o context	3/0/47	1.51	13.54
Sketch-based CoGen	27/3/20	1.81	12.98
Logits-based CoGen	32/5/13	2.30	14.18

Table 3: Human Assessments and Automated Evaluation Results (Qwen-72B/1.8B as LLM/SLM)

proving to be superior. Nonetheless, given the distinct input conditions for LLMs and SLMs, a learnable fusion model becomes essential in context-aware environments, outperforming simple pooling techniques significantly.

Logits of LLMs and SLMs. Within the fusion model, we integrate logits from LLMs and SLMs using self-adjusting weights. Figure 6 showcases the output generated by both Qwen-72B and Qwen-1.8B, employing color-coded weights to delineate their respective contributions: red signifies SLMs, blue denotes LLMs, and white represents an evenly balanced weight of 0.5. It is observed that LLMs predominantly influence the sketch of the generated content, whereas SLMs play a more significant role across the majority of tokens, underscoring the importance of collaboration.

4.6 Human Evaluation

Evaluations by LLMs have not yet been demonstrated to be infallible; consequently, we conducted human evaluations. To streamline the complexity of this assessment, annotators were instructed

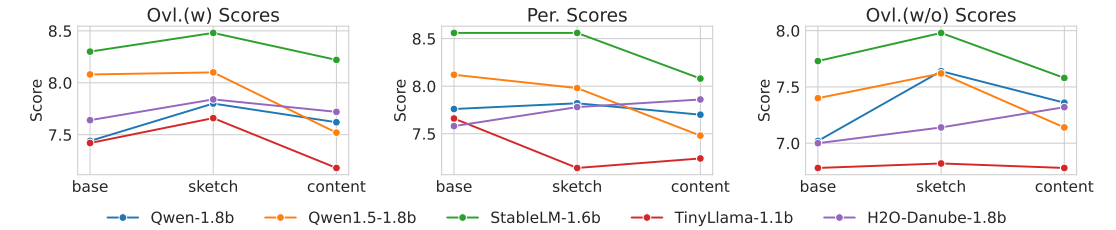


图 4: 使用草图与完整内容的 SLM 性能比较。

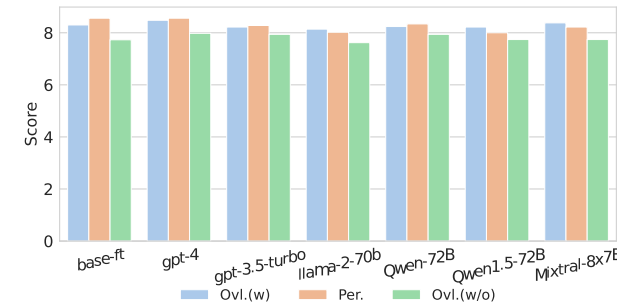


图 5: 使用由各种模型生成的草图的 StableLM 性能。

与不使用任何内容或仅使用草图相比，会降低模型性能。这种差异可以归因于潜在的冗余信息过载。如（Weston 和 Sukhbaatar）中所探讨，内容过多并不一定能提升性能，反而可能引入多余的细节。

草图的泛化能力。对于通过 GPT-4 生成的草图进行微调的小型模型，我们通过在测试中使用来自各种大型语言模型（LLM）的草图来探索其泛化潜力。具体来说，我们重点关注进行详细消融分析。图显示，相较于，使用来自其他 LLM 的草图会稍微负面影响整体和个性化评分，但在缺少上下文的情况下提升整体评分。这表明由不同模型生成的草图存在差异，并表现出有限的泛化能力。

4.5.2 基于 Logit 的 CoGenesis

Logits 融合策略。我们在中实现了一个可学习的模型，旨在合并来自上下文无关的 LLM 和包含上下文的 SLM 的 logits。此外，我们探索了简单的最大值和平均值池化策略，这些策略被认为是稳健的基线方法（Ormazabal 等，）。根据表，最大值和平均值池化方法都提升了 SLM 的性能，其中最大值池化

Metric	OvL(w)	Per.	OvL(w/o)
Models			
Qwen-72B-Chat (<i>with context</i>)	8.38	8.38	8.14
Qwen-72B-Chat (<i>w/o context</i>)	3.70	3.38	7.72
Qwen-1.8B-Chat (<i>netuned</i>)	7.44	7.76	7.02
Mean Pooling Fusing	7.76	7.84	7.42
Max Pooling Fusing	7.90	7.94	7.52
Learnable Weights Fusing	8.12	8.20	7.86

表 2: 平均融合和最大融合策略与我们的可学习融合模型的比较。

Model and Setting	Win/Tie/Lose (%)	BLEU	ROUGE-L
SLM netuned	-/50/-	2.07	13.95
LLM w/ context	38/2/10	2.61	14.66
LLM w/o context	3/0/47	1.51	13.54
Sketch-based CoGen	27/3/20	1.81	12.98
Logits-based CoGen	32/5/13	2.30	14.18

表 3: 人工评估和自动评估结果（Qwen-72B/1.8BasLLM/SLM）

证明其优越性。尽管如此，鉴于大语言模型（LLM）和小语言模型（SLM）的输入条件不同，在上下文感知环境中，可学习的融合模型变得至关重要，显著优于简单的池化技术。

大型语言模型 (LLM) 和小型语言模型 (SLM) 的输出值。在融合模型中，我们使用自适应权重整合来自 LLM 和 SLM 的输出值。图展示了由和生成的输出，使用颜色编码的权重来划分各自的贡献：红色表示 SLM，蓝色表示 LLM，白色表示均衡权重 0.5。观察可知，LLM 主要影响生成内容的轮廓，而 SLM 在大多数标记上发挥更重要的作用，凸显了协作的重要性。

4.6 人类评估

大型语言模型的评估尚未被证明是万无一失的；因此，我们进行了人工评估。为了简化这一评估的复杂性，指示了标注人员

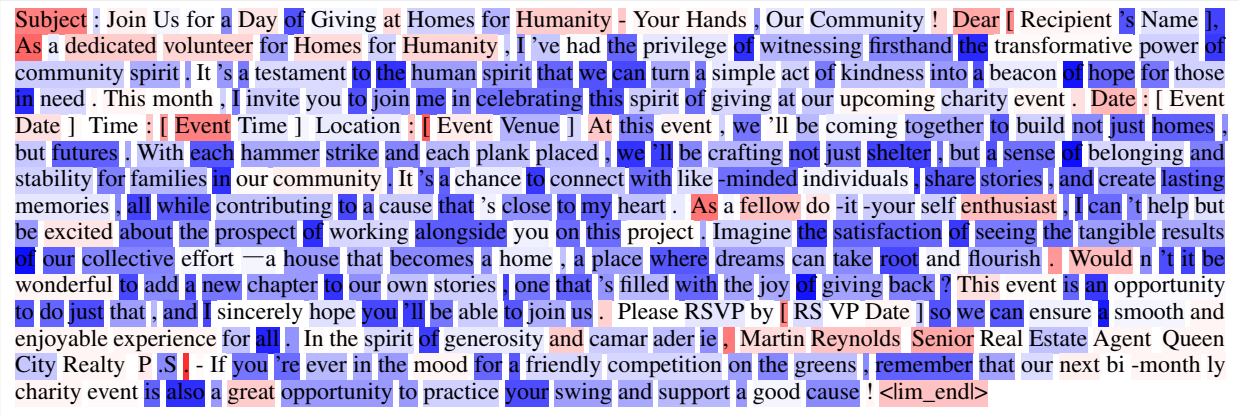


Figure 6: Visualization of LLM and SLM Logits Weights per Token During Generation. Blue signifies SLM contributions, red indicates LLM contributions, with darker shades representing higher weights.

to compare outcomes under various settings to those of the finetuned SLM, recording results as win/tie/lose, mirroring the methodology utilized in lmsys/chatbot-arena-leaderboard⁴. Additionally, we employed traditional word overlap metrics such as BLEU and ROUGE-L, calculated using the evaluate library⁵.

As shown in Table 3, the results of human evaluation are consistent with GPT-4 evaluations, where CoGenesis performs better than finetuned SLMs and LLMs without context separately, and performs closely to LLMs with context.

5 Discussion

Although our experiments have been limited to models of the same family using identical tokenizers, these methods could potentially be expanded through a tokenizer alignment strategy (Fu et al., 2023; Wan et al., 2024). This principle aligns with other logits-based decoding techniques such as speculative decoding, contrastive decoding. By aligning the tokens and probabilities of models with different tokenizers, it is feasible to facilitate knowledge transfer across various LLMs and SLMs.

Recent studies have demonstrated the potential to reconstruct prompts based on the distribution of next token tokens (Morris et al., 2023, 2024). However, the accuracy of extraction, particularly the exact match scores, remains discouragingly low. Furthermore, since only the top-k logits for each token are utilized in our experimental, the cost of reconstruction is prohibitively high. Therefore, logits-based collaboration remains sufficiently

⁴<https://huggingface.co/spaces/lmsys/chatbot-arena-leaderboard>

⁵<https://github.com/huggingface/evaluate>

Settings	OvL(w/)	Per.	OvL(w/o)
LLM w/ context	8.38	8.38	8.14
LLM w/o context	3.70	3.38	7.72
FT SLM (rk=0 toks)	7.44	7.76	7.02
LLM+SLM (rk=8 toks)	7.72	7.96	7.38
LLM+SLM (rk=16 toks)	7.78	7.84	7.22
LLM+SLM (rk=32 toks)	7.94	7.98	7.30
LLM+SLM (rk=64 toks)	7.94	8.04	7.54
LLM+SLM (rk=128 toks)	7.98	8.06	7.44
Logit-based CoGenesis	8.12	8.20	7.86

Table 4: Comparative Analysis of First Token Quantities used in Logit-based CoGenesis.

secure, and can be further enhanced with the implementation of encryption and noise addition algorithms. As shown in Table 4, we have investigated the logit-based CoGenesis approach, which enhances privacy by uploading only the previous few tokens instead of the entire response. This approach, inspired by the principle that a good start leads to effective completion (Jain et al., 2024; Wang and Zhou, 2024), suggests that initial guidance from LLMs on the first few tokens can direct SLMs to independently generate the remainder of the response. Performance data indicate improvements when transferring just 8, 16, or 32 tokens to cloud LLMs, compared to a fine-tuned SLM alone, with the entire response typically exceeding 500 tokens. Increasing the number of transferred tokens boosts the LLM + SLM performance, allowing us to balance enhanced performance against reduced privacy risks.

6 Related Works

The advent of LLMs has revolutionized the field of instruction following, with models being trained

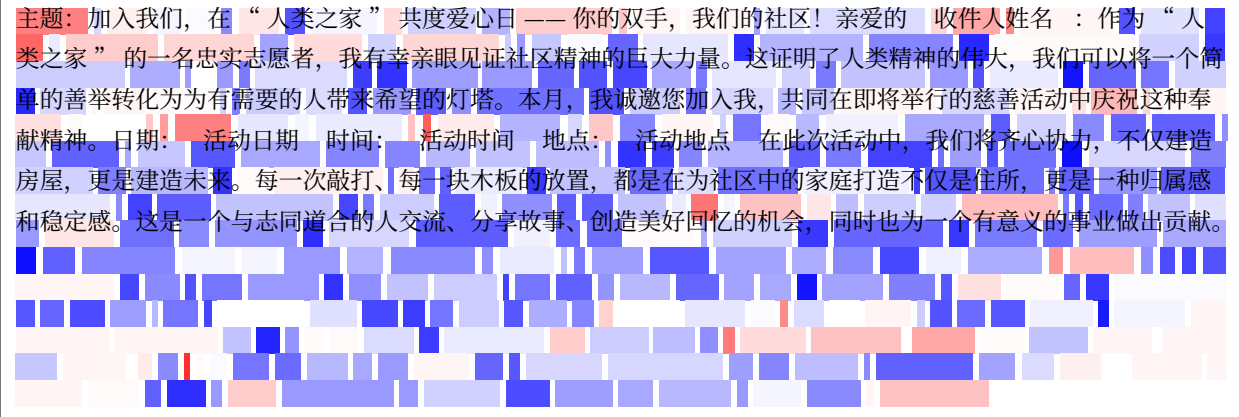


图 6: 生成过程中每个 Token 的 LLM 和 SLM Logits 权重可视化。蓝色表示 SLM 的贡献，红色表示 LLM 的贡献，颜色越深表示权重越高。

将各种设置下的结果与微调后的 SLM 进行比较，记录结果为赢 / 平 / 输，模仿 lmsys/chatbot-arena-leaderboard 中使用的方法。此外，我们使用传统的词汇重叠指标，如 BLEU 和 ROUGE-L，通过 evaluate 库计算。

如表 所示，人类评估结果与 GPT-4 评估结果一致，其中 CoGenesis 的表现优于未微调的 SLM 和无上下文 LLM，并且与有上下文的 LLM 表现接近。

5 讨论

虽然我们的实验仅限于使用相同分词器的同一系列模型，但这些方法有可能通过分词器对齐策略扩展（Fu 等人，；Wan 等人，）。这一原理与其他基于 logits 的解码技术相一致，例如推测解码、对比解码。通过对齐具有不同分词器的模型的 tokens 和概率，有可能在各种 LLM 和 SLM 之间促进知识转移。

最近的研究表明，有可能根据下一个词元的分布来重建提示（Morris 等）。然而，提取的准确性，尤其是精确匹配分数，仍然令人沮丧地低。此外，由于在我们的实验中仅使用了每个词元的前 k 个 logits，重建的成本高得令人望而却步。因此，基于 logits 的协作仍然足够

⁴<https://huggingface.co/spaces/lmsys/chatbot-arena-leaderboard>

⁵<https://github.com/huggingface/evaluate>

Settings	OvL(w/)	Per.	OvL(w/o)
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LLM+SLM (rk=64 toks)	7.94	8.04	7.54
LLM+SLM (rk=128 toks)	7.98	8.06	7.44
Logit-based CoGenesis	8.12	8.20	7.86

表 4: 基于 Logit 的 CoGenesis 中使用的首个标记数量的比较分析。

安全，并可以通过实施加密和噪声添加算法进一步增强。如表 所示，我们研究了基于 logit 的 CoGenesis 方法，该方法通过仅上传前几个 token 而非整个响应来增强隐私。这种方法受“良好的开端促进有效完成”原则的启发（Jain 等 王和周），表明来自大型语言模型（LLM）对前几个 token 的初始指导可以引导小型语言模型（SLM）独立生成剩余的响应。性能数据显示，当仅将 8、16 或 32 个 token 传输到云端 LLM 时，相比单独微调的 SLM，性能有所提升，而整个响应通常超过 500 个 token。增加传输的 token 数量可以提高 LLM + SLM 的性能，使我们能够在增强性能与重新

6 相关工作

大型语言模型的出现已经彻底改变了指令执行领域，模型正在被训练

on diverse and complex instruction sets, enabling them to perform a wide array of tasks from creative writing (Franceschelli and Musolesi, 2023) to coding (Qian et al., 2023) and debugging (Jimenez et al., 2023). The push towards collecting high-quality instruction data (Wang et al., 2023c; Ding et al., 2023; Xu et al., 2023a) has allowed for the development of both proprietary and open-source medium-scale language models adept at following instructions. However, the reliance on cloud-based proprietary models like ChatGPT and GPT-4 for instruction execution raises significant privacy concerns due to the potential risks associated with uploading sensitive data (Achiam et al., 2023; Team et al., 2023; Liu et al., 2023). To mitigate these risks, various privacy-preserving techniques have been employed, albeit with limitations in completely securing user data (Cummings et al., 2023; Kan et al., 2023; Chen et al., 2023b; Wu et al., 2023). Furthermore, there is a notable gap in instruction datasets, particularly in the inclusion of contextual information (Salemi et al., 2023; Wang et al., 2023b), highlighting an area for further exploration in context-aware instruction formulation to enhance privacy and personalization.

In parallel, the exploration of mixed-scale model collaboration emerges as a promising avenue to address the scalability, efficiency (Xia et al., 2024), and privacy (Yao et al., 2023) challenges inherent to LLMs. While larger models benefit from increased capabilities, their high inference costs and privacy concerns contrast with the lower costs and greater accessibility of smaller models (Bai et al., 2023; Gunasekar et al., 2023; Zhang et al., 2024; Grangier et al., 2024; Singer et al., 2024). Research in this domain is bifurcated into collaborative training and inference strategies, including offsite-tuning (Xiao et al., 2023) and speculative decoding (Leviathan et al., 2023), aiming to leverage the strengths of both large and small models (Mitchell et al., 2023; Liu et al., 2024). This paper specifically investigates collaborative inference methods, including sketch-based and logit-based approaches, to enhance the efficiency and privacy of LLMs, suggesting a promising direction in utilizing mixed-scale models for instruction following tasks.

7 Conclusion

This paper investigates context-aware instruction following, enriching prompts with detailed user privacy information. We outline a pipeline for

creating context-aware instructions. To mitigate privacy issues, we introduce CoGenesis, a collaborative framework between SLMs and LLMs utilizing sketches and logits. Our results highlight the advantages of mixed-scale model collaboration, suggesting fruitful directions for future research.

Limitations

This study explores context-aware instruction following, introducing strategies for collaboration between large and small models to address privacy concerns. We developed a synthetic dataset for context-aware instruction following to empirically test our approaches. Our findings suggest that this model collaboration can significantly mitigate privacy risks associated with using public API-based LLMs. However, our dataset is limited in size and was specifically crafted for preliminary validation. Future work will focus on expanding this dataset to enhance its quality, realism, and diversity. Additionally, our proposed methods, particularly the logits-based approach, are currently restricted to models sharing the same tokenizer. Further research on tokenizer alignment is necessary to broaden the applicability of our strategies.

Ethics Statement

The advent of LLMs has underscored the urgent need to address privacy and security concerns within the realm of artificial intelligence. This paper concentrates on the privacy challenges posed by context-aware instruction-following applications of LLMs, proposing methods to mitigate these concerns without compromising the models’ effectiveness. We emphasize the ethical imperative of protecting user data, adopting a strategy that involves generating synthetic datasets using GPT-4 for training and testing. This approach ensures that our research does not compromise real-world user privacy by preventing any potential data leakage. In essence, our work not only seeks to advance the technological capabilities of LLMs but also to uphold the highest standards of ethical responsibility by safeguarding user privacy through innovative and secure data handling practices.

Acknowledgements

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在多样且复杂的指令集上，使其能够执行从创作写作（Franceschelli 和 Musolesi, 2023）到编码（Qian 等, 2023）和调试（Jimenez 等, 2023）的各种任务。对高质量指令数据的收集推动（Wang 等, 2023c；Ding 等, 2023；Xu 等, 2023a）促成了专有和开源中等规模语言模型的发展，这些模型善于遵循指令。然而，依赖诸如 ChatGPT 和 GPT-4 的云端专有模型来执行指令引发了显著的隐私问题，因为上传敏感数据可能带来潜在风险（Achiam 等, 2023；Team 等, 2023；Liu 等, 2023）。为缓解这些风险，各种隐私保护技术被应用，尽管在完全保护用户数据方面存在局限性（Cummings 等, 2023；Kan 等, 2023；Chen 等, 2023b；Wu 等, 2023）。此外，在

与此同时，混合规模模型协作的探索成为应对大型语言模型（LLMs）固有的可扩展性、效率（Xia 等, 2024）和隐私（Yao 等, 2023）挑战的一个有前景的途径。虽然更大的模型因能力增强而受益，但其高推理成本和隐私问题与较小模型的低成本和更高可访问性形成对比（Bai 等, 2023；Gunasekar 等, 2023；Zhang 等, 2024；Grangier 等, 2024；Singer 等, 2024）。该领域的研究分为协作训练和推理策略，包括离线调优（Xiao 等, 2023）和推测性解码（Leviathan 等, 2023），旨在利用大模型和小模型的优势（Mitchell 等, 2023；Liu 等, 2024）。本文特别研究协作推理方法，包括基于草图和基于 Logit 的方法，以提高 LLMs 的效率和隐私保护，提出了一个有前景的方向。

7 结论

本文研究了上下文感知的指令执行，通过提供详细的用户隐私信息来丰富提示。我们概述了一个用于

创建上下文感知的指令。为了减轻隐私问题，我们引入了 CoGenesis，这是一个使用草图和对数几率的 SLM 与 LLM 之间的协作框架。我们的结果强调了混合规模模型协作的优势，表明了未来研究的有益方向。

限制

本研究探讨了上下文感知的指令跟随，介绍了大型模型与小型模型之间协作的策略，以解决隐私问题。我们开发了一个用于上下文感知指令跟随的合成数据集，以实证测试我们的方法。我们的研究表明，这种模型协作可以显著减轻使用基于公共 API 的大语言模型时的隐私风险。然而，我们的数据集规模有限，并且是专门为初步验证而制作的。未来的工作将侧重于扩展该数据集，以提高其质量、真实性和多样性。此外，我们提出的方法，特别是基于 logits 的方法，目前仅限于共享相同分词器的模型。还需要在分词器对齐方面进行进一步研究，以扩大我们策略的适用性。

伦理声明

大型语言模型（LLM）的出现凸显了在人工智能领域中迫切需要解决隐私和安全问题。本文集中讨论了由 LLM 的上下文感知指令执行应用所带来的隐私挑战，并提出了在不影响模型有效性的前提下缓解这些问题的方法。我们强调保护用户数据的伦理必要性，采取的策略包括使用 GPT-4 生成合成数据集进行训练和测试。这种方法确保我们的研究不会通过潜在的数据泄露而影响现实世界的用户隐私。本质上，我们的工作不仅旨在提升 LLM 的技术能力，同时通过创新和安全的数据处理实践维护用户隐私，从而坚持最高的伦理责任标准。

致谢

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A Dataset Details

A.1 Context-Aware Instructions

The four steps for constructing context-aware instructions are as follows.

Group Portraits. We begin by constructing highly diverse user group portraits from the real world, encompassing a wide range of groups such as college students, programmers, and various other professions. For each group, we define their age range, identify their professional field or occupation, and enumerate typical activities and hobbies to capture the group’s unique interests. Additionally, we delineate specific scenarios in which these groups might utilize an AI assistant for personalized and creative text generation in both their professional and personal lives. Examples of such use cases include drafting business emails, writing creative blogs, composing academic papers, and crafting extended tweets.

User Profile. Building upon our diverse user group portraits, we next develop individual user profiles with rich detail. Emphasizing consistency and realism, our process involves four steps: 1) Personal Writing Style: Tailoring language use and expression unique to each character. 2) Private Information: Creating 5-10 fictional details for each profile, including life events and technology interactions. 3) Smart Device Usage: Generating 5-10 fictional activity logs per profile, covering messages, purchases, schedules, and more. Our aim is to shape distinct, multi-dimensional characters for a variety of AI writing applications.

Writing Task Instructions. In this phase, we craft text creation tasks tailored to our user characters, ensuring alignment with their professions, hobbies, and lifestyles. These tasks are intricately linked to their mobile phone activity logs and personal details, weaving the characters’ experiences, social media activities, and AI assistant interactions into the narratives. For each character, we develop K tasks that are both realistic and contextually relevant.

Personalized Generations. Here, we produce personalized texts adhering to specific guidelines. Our focus is on crafting authentic and coherent narratives that vividly reflect each character’s professional and personal life. By adapting to the user’s unique writing style, we aim to create personalized and stylistically distinctive content. To ensure coherence and relevance across various tasks, content for each task is generated sequentially.

Following these four steps, we utilize GPT-4 to create instructions with user context and responses for subsequent experiments.

A.2 LaMP Dataset Processing

To minimize the personalization gap between the target content and user profiles in the Avocado Research Email and Citation Network Papers datasets in LaMP⁶, we have implemented a two-step processing approach. Initially, we retrieve content that most closely aligns with the user profile using `intfloat/e5-mistral-7b-instruct` embeddings (Wang et al., 2023a). Subsequently, we encode the target content and the entire profile using style embeddings (Wegmann et al., 2022; Patel et al., 2023), selecting only the most personalized samples. Furthermore, the content lengths and profiles of samples in LaMP vary significantly, ranging from 10 to 100,000 tokens⁷, exceeding the context length capabilities of contemporary models. Consequently, we have selected samples with a minimum length of 128 characters for paper abstracts and 64 for email bodies, and a maximum length of 1024 characters for both. Owing to the unavailability of a test dataset in LaMP, we repurposed the dev dataset as our test set. Additionally, we divided the filtered training data into actual training and validation datasets, using a 9:1 split ratio.

A.3 Dataset Statistics

The statistical results of the final processed dataset are presented in Table 1.

Dataset	Context-aware	Avocado Emails	Paper Abstracts
Total Users	1736	N/A	N/A
Avg Profile Length	1182	618	1000
Output Length	155	178	158
Train Samples	1346	1,137	1,448
Dev Samples	150	127	161
Test Samples	240	346	357

Table 5: The table presents the statistics of constructed dataset and public datasets.

Our synthetic instruction datasets include a variety of instruction types, such as preparing a speech, designing a plan, and more. All of these tasks require models to utilize personal context information, including previous activities and schedules. To illustrate, we display the top 10 most common root verbs and top 10 direct noun objects in our constructed datasets as shown in Figure 6

⁶<https://lamp-benchmark.github.io/download>

⁷<https://github.com/openai/tiktoken/>

数据集详情

A.1 上下文感知指令

构建上下文感知指令的四个步骤如下。

群体画像。我们首先从现实世界构建高度多样化的用户群体画像，涵盖大学生、程序员以及各种其他职业等广泛群体。对于每个群体，我们定义其年龄范围，确定其专业领域或职业，并列举典型活动和爱好，以捕捉该群体的独特兴趣。此外，我们还描述这些群体可能在何种具体场景中使用 AI 助手进行个性化和创意文本生成，无论是在工作还是个人生活中。这类使用场景的示例包括撰写商务邮件、撰写创意博客、撰写学术论文以及创作长篇推文。

用户档案。在建立了多样化的用户群体画像的基础上，我们接下来开发具有丰富细节的个人用户档案。强调一致性和真实性，我们的流程包括四个步骤：1) 个人写作风格：针对每个角色量身定制独特的语言使用和表达方式。2) 私人信息：为每个档案创建 5-10 个虚构细节，包括生活事件和技术互动。3) 智能设备使用：为每个档案生成 5-10 条虚构活动记录，涵盖消息、购买、日程等。我们的目标是为各种 AI 写作应用塑造独特、多维的角色。

写作任务说明。在此阶段，我们为角色设计文本创作任务，确保与他们的职业、爱好和生活方式保持一致。这些任务与他们的手机活动记录和个人信息密切相关，将角色的经历、社交媒体活动和 AI 助手互动融入叙事中。对于每个角色，我们开发一个既真实又符合情境的任务。

个性化生成。在这里，我们根据特定的指导方针生成个性化文本。我们的重点是创作真实且连贯的叙述，生动地反映每个角色的职业和个人生活。通过适应用户独特的写作风格，我们旨在创造个性化且具有风格特色的内容。为了确保在各项任务中的连贯性和相关性，每项任务的内容都是按顺序生成的。

按照这四个步骤，我们利用 GPT-4 创建带有用户上下文和响应的指令，以用于后续实验。

A.2 LaMP 数据集处理

为了在 LaMP 6 中的 Avocado Research Email 和 Citation Network Papers 数据集里最小化目标内容与用户简介之间的个性化差距，我们实现了一个两步处理方法。首先，我们使用 `intfloat/e5-mistral-7b-instruct` 嵌入（Wang 等，）检索与用户简介最匹配的内容。随后，我们使用风格嵌入（Wegmann 等，；Patel 等，）对目标内容和整个简介进行编码，仅选择最具个性化的样本。此外，LaMP 中样本的内容长度和简介差异显著，范围从 10 到 100,000 个标记，超过了现代模型的上下文长度能力。因此，我们选择了论文摘要长度最少为 128 个字符、电子邮件正文长度最少为 64 个字符，且两者的最大长度均为 1024 个字符的样本。

A.3 数据集统计

最终处理数据集的统计结果列于表。

Dataset	Context-aware	Avocado Emails	Paper Abstracts
Total Users	1736	N/A	N/A
Avg Profile Length	1182	618	1000
Output Length	155	178	158
Train Samples	1346	1,137	1,448
Dev Samples	150	127	161
Test Samples	240	346	357

Table 5: The table presents the statistics of constructed dataset and public datasets.

我们的合成指令数据集包括多种类型的指令，例如准备演讲、设计计划等。所有这些任务都要求模型利用个人上下文信息，包括以前的活动和日程安排。为说明这一点，我们展示了在构建的数据集中最常见的前 10 个根动词和前 10 个直接名词对象，如图所示

⁶<https://lamp-benchmark.github.io/download>

⁷<https://github.com/openai/tiktoken/>

Verb	Percent (%)	Verb	Percent (%)
write	19.4	post	24.0
draft	16.0	article	13.8
compose	13.9	speech	12.5
create	11.5	proposal	10.2
develop	8.7	guide	7.2
prepare	8.2	series	4.4
craft	6.3	piece	4.7
curate	4.5	plan	3.2
design	2.1	outline	3.0
script	2.0	email	2.8

Table 6: The top 10 most common root verbs and top 10 direct noun objects in our constructed datasets.

B Model Details

For all open-source Large Language Models, we utilize vLLM⁸ for efficient inference, setting the temperature to 0.7, top-p to 0.9, and the maximum number of new tokens to 1024.

Regarding the specialization of Small Language Models (SLMs), we apply a range of learning rates {5e-6, 8e-6, 1e-5, 2e-5, 5e-5} across different models and datasets. Furthermore, we implement an early stopping strategy to identify the optimal model based on validation performance as the specialized models. We fine-tune each model using a batch size of 8, max sequence length of 4096, across four A6000 48GB GPUs.

For the combined model, we employ a three-layer neural network featuring ReLU activation, with sigmoid activation applied to determine the final weights. At each generation step, only the top 10 logits from both LLMs and SLMs are utilized. The intermediate hidden layers are configured with sizes of 512 and 16, respectively. The model is trained using a learning rate of 2e-3 and a batch size of 2. The combined model is trained on the training dataset, with both LLMs and SLMs assigned the same target response. Additionally, an early stopping strategy based on the validation set performance is employed to select the optimal combined model. For all the aforementioned experiments, we calculate the mean scores using three distinct random seeds.

C Evaluation Details

Owing to the costs associated with evaluation, we assess only a portion of the test samples. To account for GPT-4’s evaluation stability, we plot a curve illustrating the relationship between scores and the number of evaluated samples. As depicted

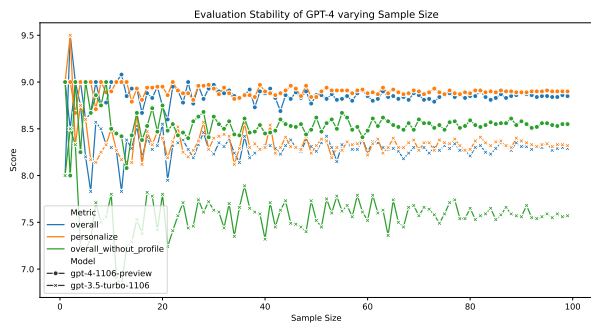


Figure 7: This illustration demonstrates evaluation consistency and stability of GPT-4 as a judge.

in Figure 7, the results stabilize once the number of samples reaches 100. Therefore, we randomly select 100 samples from the test set, of which only 80 samples are utilized in the widely recognized benchmark, MT-Bench⁹.

D Prompt Details

Prompts designed for querying Large Language Models (LLMs) both with and without context are outlined in Table 7. Prompts intended for extracting outlines are illustrated in Table 8. Prompts used for GPT-4 based evaluation are depicted in Table 9.

E Related Works

E.1 Instruction Following and Privacy

Large language models (LLMs), after being trained on high-quality instruction data and calibrated to align with human intentions, have acquired the capability to execute instructions across a range of activities such as creative writing (Franceschelli and Musolesi, 2023), coding (Qian et al., 2023), debugging (Jimenez et al., 2023), and various other text-based tasks (Bubeck et al., 2023). Contemporary research in instruction following prioritizes the acquisition of high-quality data (Wang et al., 2023c), which encompasses instructions of varied complexity (Xu et al., 2023a) and diversity (Ding et al., 2023; Cui et al., 2023), as well as ensuring a minimal dataset size for effective generalization (Chen et al., 2023a; Wei et al., 2023). Thanks to these technological advancements along with alignment algorithms (Rafailov et al., 2023), medium-scale language models with around ten billion parameters have been made open-source and perform adeptly at following instructions (Tunstall et al., 2023; Jiang et al., 2023; Zhang et al., 2024).

Verb	Percent (%)	Verb	Percent (%)
write	19.4	post	24.0
draft	16.0	article	13.8
compose	13.9	speech	12.5
create	11.5	proposal	10.2
develop	8.7	guide	7.2
prepare	8.2	series	4.4
craft	6.3	piece	4.7
curate	4.5	plan	3.2
design	2.1	outline	3.0
script	2.0	email	2.8

Table 6: The top 10 most common root verbs and top 10 direct noun objects in our constructed datasets.

B 型号详情

对于所有开源的大型语言模型，我们使用 vLLM 进行高效推理，设置温度为 0.7，top-p 为 0.9，最大新生成的 token 数为 1024。

关于小型语言模型（SLMs）的专门化，我们在不同的模型和数据集上应用了一系列学习率 {5e-6, 8e-6, 1e-5, 2e-5, 5e-5}。此外，我们实施了早停策略，根据验证性能确定最佳模型作为专门化模型。我们使用批量大小为 8、最大序列长度为 4096，在四块 A6000 48GB GPU 上对每个模型进行微调。

对于组合模型，我们采用一个三层神经网络，使用 ReLU 激活函数，并应用 sigmoid 激活函数来确定最终权重。在每个生成步骤中，仅使用来自 LLM 和 SLM 的前 10 个 logits。中间隐藏层的大小分别配置为 512 和 16。模型使用学习率 2e-3 和批量大小 2 进行训练。组合模型在训练数据集上进行训练，LLM 和 SLM 均分配相同的目标响应。此外，还采用基于验证集性能的提前停止策略来选择最佳组合模型。对于上述所有实验，我们使用三个不同的随机种子计算平均得分。

C 评估详情

由于评估相关的成本，我们仅评估部分测试样本。为了考虑 GPT-4 评估的稳定性，我们绘制了一条曲线，说明评分与评估样本数量之间的关系。如图所示

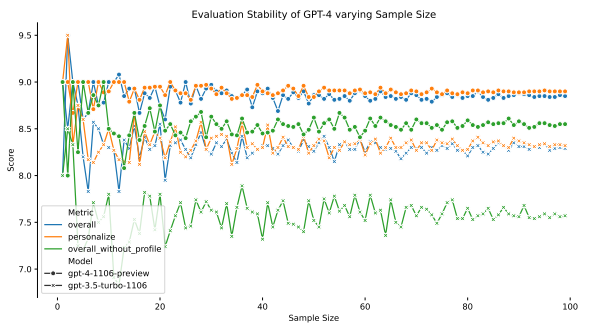


图 7: 此图说明了 GPT-4 作为评审的评估一致性和稳定性。

在图 中，一旦样本数量达到 100，结果就会稳定。因此，我们从测试集中随机选择了 100 个样本，其中只有 80 个样本被用于广泛认可的基准测试 MT-Bench。

D 提示详情

用于查询大型语言模型（LLM）的提示，包括有上下文和无上下文的提示，列在表 中。用于提取大纲的提示示例见表 中。用于基于 GPT-4 的评估的提示如表 所示。

相关工作

E.1 遵循指令与隐私

大型语言模型（LLM）在经过高质量指令数据的训练并校准以符合人类意图后，已经获得了在一系列活动中执行指令的能力，例如创意写作（Franceschelli 和 Musolesi, 2023）、编程（Qian 等, 2023）、调试（Jimenez 等, 2023）以及各种其他基于文本的任务（Bubeck 等, 2023）。当代关于指令遵循的研究优先考虑高质量数据的获取（Wang 等, 2023c），这些数据涵盖了不同复杂度（Xu 等, 2023a）和多样性（Ding 等, 2023；Cui 等, 2023）的指令，同时确保最小化数据集规模以实现有效的泛化（Chen 等, 2023a；Wei 等, 2023）。得益于这些技术进步以及对齐算法（Rafailov 等, 2023），具有约一百亿参数的中等规模语言模型已被开源，并且在执行指令方面表现出色。

⁸<https://github.com/vllm-project/vllm>

⁹<https://huggingface.co/spaces/lmsys/mt-bench>

⁸<https://github.com/vllm-project/vllm>

⁹<https://huggingface.co/spaces/lmsys/mt-bench>

Nevertheless, premier chat models like ChatGPT, GPT-4, Claude, and Gemini remain proprietary, largely due to commercial considerations, necessitating that our instruction data be uploaded to the cloud (Achiam et al., 2023; Team et al., 2023). While these models serve as potent AI assistants in daily professional and personal endeavors, they also pose significant privacy risks (Liu et al., 2023). To address privacy concerns, LLMs have employed various techniques (Cummings et al., 2023), including text sanitization (Kan et al., 2023), differential privacy (Chen et al., 2023b; Wu et al., 2023), and hidden representations (Zhou et al., 2022, 2023). However, these methods still involve uploading potentially sensitive data to the cloud, which inherently cannot eliminate the risk of privacy breaches.

Moreover, current instruction datasets predominantly cover general domains, with insufficient focus on contextual information modeling. The most closely related works involve personalized response generation, evolving from traditional benchmarks (Salemi et al., 2023; Wang et al., 2023b). The integration of extensive context information into general open-domain instructions remains an area of ongoing exploration. This paper aims to delve into context-aware instruction formulation as a means to advance research on privacy considerations within the realm of instruction following.

E.2 Mixed-Scale Models Collaboration

The “scaling law” in language modeling posits that models with a greater number of parameters exhibit enhanced capabilities (Kaplan et al., 2020). However, these more robust models also encounter challenges related to higher inference costs, efficiency (Xia et al., 2024), and privacy concerns (Yao et al., 2023). Conversely, smaller models, ranging from 1 to 2 billion parameters, are gaining popularity due to their increasingly impressive performance (Bai et al., 2023; Gunasekar et al., 2023; Zhang et al., 2024; Grangier et al., 2024; Singer et al., 2024). These specialized models are coupled with lower inference costs and the feasibility of deployment on consumer-grade desktops and smartphones (MLC team, 2023). Collaborations between mixed-scale models represent a promising research avenue. The body of current research in this area primarily falls into two categories: training and inference. For collaborative training, Offsite-tuning has been introduced as a method to protect both user data and the privacy of large models (Xiao et al., 2023; Zhang et al., 2023c). This approach

involves using an emulator derived from LLMs, fine-tuning it on specific downstream data, and subsequently integrating the learned parameters back into the LLMs. On the inference front, techniques like speculative decoding (Leviathan et al., 2023; Xia et al., 2024) and contrastive decoding (Li et al., 2022; O’Brien and Lewis, 2023) aim to enhance and expedite LLMs’ inference processes by leveraging smaller draft or expert models. Additionally, emulator tuning (Mitchell et al., 2023) and proxy tuning (Liu et al., 2024) have been devised to economize on fine-tuning large models; however, they can also be considered forms of collaborative decoding during inference. This paper focuses on examining collaboration during inference, specifically investigating sketch-based and logit-based methods.

尽管如此，像 ChatGPT、GPT-4、Claude 和 Gemini 这样的顶级聊天模型仍然是专有的，这在很大程度上是出于商业考虑，因此我们的指令数据必须上传到云端 (Achiam 等 Team 等)。虽然这些模型在日常专业和个人事务中作为强大的人工智能助手，但它们也带来了显著的隐私风险 (Liu 等)。为了解决隐私问题，大型语言模型采用了各种技术 (Cummings 等)，包括文本清理 (Kan 等)、差分隐私 (Chen 等 Wu 等) 和隐藏表示 (Zhou 等 <b67)

此外，当前的指令数据集主要涵盖通用领域，对上下文信息建模的关注不足。最相关的工作涉及个性化响应生成，其发展源自传统基准 (Salemi 等, 2023; Wang 等, 2023b)。将大量上下文信息整合到通用开放域指令中仍然是一个持续探索的领域。本文旨在探讨上下文感知的指令制定，以推动在指令遵循领域中对隐私考虑的研究。

E.2 混合规模模型合作

语言模型中的“规模定律”假设参数数量更多的模型具有更强的能力 (Kaplan 等, 2020)。然而，这些更强大的模型也面临与更高的推理成本、效率 (Xia 等, 2024) 以及隐私问题 (Yao 等, 2023) 相关的挑战。相反，参数数量在 10 亿到 20 亿之间的小型模型由于其越来越令人印象深刻的性能而受到欢迎 (Bai 等, 2023; Gunasekar 等, 2023; Zhang 等, 2024; Grangier 等, 2024; Singer 等, 2024)。这些专用模型具有更低的推理成本，并且可以在消费者级桌面电脑和智能手机上部署 (MLC 团队, 2023)。混合规模模型之间的协作代表了一条有前景的研究途径。目前该领域的研究主要分为两类：训练和推理。在协作训练方面，已经提出了离线微调 (Offsite-tuning) 作为一种方法来原型化

涉及使用源自大语言模型 (LLMs) 的模拟器，对其在特定下游数据上进行微调，并将学到的参数整合回 LLMs。在推理方面，诸如推测解码 (Leviathan 等 Xia 等) 和对比解码 (Li 等 O’Brien 和 Lewis) 等技术旨在通过利用较小的草稿模型或专家模型来增强和加速 LLMs 的推理过程。此外，模拟器微调 (Mitchell 等) 和代理微调 (Liu 等) 已被设计用于节省大模型的微调成本；然而，它们也可以被视为推理过程中的一种协作解码形式。本文重点研究推理过程中的协作，具体

SYSTEM PROMPT FOR REQUEST WITHOUT CONTEXT You are now a helpful personal AI assistant. Aim for insightful and high-quality solutions that make users satisfied.
SYSTEM PROMPT FOR REQUEST WITH CONTEXT IN CONTEXT-AWARE You are now a helpful personal AI assistant. You should emulate the author’s style and tone based on provided history content. Your responses should be detailed and informative, using the personal information reasonably in the user’s profile. Aim for insightful and high-quality solutions that make users satisfied.
SYSTEM PROMPT FOR REQUEST WITH CONTEXT IN PERSONALIZED EMAILS AND PAPERS You are now a helpful personal AI assistant. You should emulate the author’s style and tone based on provided history content. Your responses should be detailed and informative, matching the author’s unique writing approach. Aim for insightful and high-quality solutions that make users satisfied.
<u>FEW-SHOT INSTRUCTION FOR REQUEST WITH CONTEXT IN CONTEXT-AWARE</u> ## User Profile {profile} ## User Writing History {history} ## Task {task}
<u>FEW-SHOT INSTRUCTION FOR REQUEST WITHOUT CONTEXT IN CONTEXT-AWARE</u> {task}
<u>FEW-SHOT INSTRUCTION FOR REQUEST WITH CONTEXT IN PERSONALIZED EMAILS</u> ## History Emails {examples} ## Task Compose an email for the subject ‘{task}’ that matches the author’s unique style and tone.
<u>FEW-SHOT INSTRUCTION FOR REQUEST WITHOUT CONTEXT IN PERSONALIZED EMAILS</u> Compose an email for the subject ‘{task}’
<u>FEW-SHOT INSTRUCTION FOR REQUEST WITH CONTEXT IN PERSONALIZED PAPERS</u> ## History Paper Abstracts {examples} ## Task Compose an abstract for the title ‘{task}’ that matches the author’s unique content, style and tone.
<u>FEW-SHOT INSTRUCTION FOR REQUEST WITHOUT CONTEXT IN PERSONALIZED PAPERS</u> Compose an abstract for the title ‘{task}’

Table 7: Prompts for querying LLMs with and without context.

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<u>FEW-SHOT INSTRUCTION FOR REQUEST WITH CONTEXT IN CONTEXT-AWARE</u> ## User Profile {profile} ## User Writing History {history} ## Task {task}
<u>FEW-SHOT INSTRUCTION FOR REQUEST WITHOUT CONTEXT IN CONTEXT-AWARE</u> {task}
<u>FEW-SHOT INSTRUCTION FOR REQUEST WITH CONTEXT IN PERSONALIZED EMAILS</u> ## History Emails {examples} ## Task Compose an email for the subject ‘{task}’ that matches the author’s unique style and tone.
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<u>FEW-SHOT INSTRUCTION FOR REQUEST WITHOUT CONTEXT IN PERSONALIZED PAPERS</u> Compose an abstract for the title ‘{task}’

表 7：带上下文和不带上下文查询大型语言模型的提示。

